

Book III - The Spiral Steward

Part 4 - Living Algorithms

Chapter 15 - Why Genome Browsers Are Hard to Use

BRIDGE 8 — Why Genome Browsers Are Hard to Use

(And What That Tells Us About How We Think About Life)

Type: Tool critique + living systems philosophy + LivingWorks vision **Core Claim:** The difficulty of the genome browser interface is not a minor UX problem. It is a symptom of the machine model of biology — treating life as a linear sequence of parts rather than a living network of relationships. A tool built on the living model would feel like navigating a landscape you already half-know.

The Argument in One Line

The interface reflects the philosophy. If your tool is hard to use, ask what assumption about the subject made it that way.

The genome browser is hard to use because it was built by people who think of the genome as a catalogue — a linear sequence of parts at fixed addresses. Navigate by coordinate. `chr6:31,000,000`. That is not a place. It is an address in a system that only experts can read.

The genome is actually a living network. A tool built on that truth would navigate differently.

What Makes Genome Browsers Hard

You must know before you can find. There is no wandering. No discovery. No following a thread of curiosity. You arrive knowing your destination. You confirm what you expected. You leave. No genome

browser has ever surprised its user with something they didn't already suspect.

Scale jumps are disorienting. Whole genome to single gene to individual base pair — teleportation, not travel. You lose context. You forget why you came.

Everything is a snapshot. The genome is always in process. The browser freezes it. A living system in a static frame is not quite alive anymore.

The phenotype is invisible. You see sequence. You see expression levels. You never see what the organism does or becomes. The most important question — *what does this produce in the world* — is unanswerable from inside the browser.

Relationships are hidden. The genome's defining feature is that everything regulates everything else. The browser shows a line. The network — the actual intelligence — doesn't appear.

The Root Cause

The genome browser was built on the machine model. It treats the genome as a linear sequence of parts to be catalogued and addressed.

The genome is a living network to be understood.

The wrong mental model produced the wrong interface. The interface difficulty is the proof.

What the Right Interface Would Feel Like

Start with the organism. Drill down to the genome. Not: here is the sequence, infer the biology. But: here is the biology — the cell, the tissue, the phenotype — now descend to the code that produces it.

Navigate by question, not coordinate. Not: `chr17:43,044,294` But: *show me where breast cancer risk is encoded*. Arrive with context. Know why you're there. See what it does in the world.

Navigate by relationship, not position. The gene regulatory network as the primary view. Follow edges inward to find what controls something. Follow edges outward to find what something affects. Move through the logic of life, not through coordinates.

Time as a navigation dimension. Watch the network change through developmental time, response time, evolutionary time. The cell becoming a neuron — visible as a changing pattern of activation, not as a data table.

The system narrates. As you explore, the browser tells you what you're looking at. Not a tooltip — a guide. Building understanding as you move, not requiring it before you start.

The Test of a Living Tool

Does the tool feel like the thing it is working with?

A tool built on the machine model of life feels mechanical — rigid, coordinate-based, expert-only, snapshot-frozen.

A tool built on the living model of life feels like navigating a landscape you already half-know. Relational. Time-aware. Question-driven. Connected from sequence to phenotype, from code to consequence.

That is the test. LivingWorks passes it. The current genome browser does not.

The LivingWorks Connection

The genome browser redesigned is not a separate project from LivingWorks.

It is the same interface applied to biological systems instead of designed ones. The same navigation philosophy:

- Start with the living thing, not its coordinates
- Navigate by relationship and question
- Show time always
- Connect code to what it produces in the world

LivingWorks for sites and communities. The redesigned genome browser for cells and genomes. The same tool. The same philosophy. Two scales of the same living system.

Cold Open:

"Try to look something up in a genome browser. First you'll need the chromosome number. Then the start position. Then the end position. If you don't already know those — you can't even begin. That's not a tool that wants to be understood. That's a tool designed by people who already understand it for people who already understand it. And it tells you everything about how biology thinks about itself."

The Difficulty Is the Argument: Walk through exactly what makes genome browsers hard. Show that each difficulty is a direct consequence of the machine model.

What the Living Model Would Produce: Navigate by question. Navigate by relationship. Show time. Connect to phenotype. The interface that the biology actually calls for.

LivingWorks: The same navigation philosophy applied to designed living systems. One interface for reading life's code — biological or built.

The Closing:

"Every tool encodes a philosophy. The genome browser encodes the philosophy that life is a catalogue. We are building tools that encode a different philosophy: that life is a network. That it moves through time. That it cannot be understood in isolation from what it produces. When the tools change, the science changes. When the science changes, the civilization changes."

The Living Age Trilogy